Interpretable Spatial-Temporal Video Transformer (ISTVT) for Deepfake Detection

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Project Report

Course: EE565

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1. Abstract

With the fast growth of Deepfake technology, our personal privacy and information security are under serious threat. To build reliable Deepfake detection systems, researchers have started using both spatial and temporal features in videos, applying methods like recurrent neural networks (RNNs) and 3D convolutional neural networks (3D CNNs). However, these models still have limitations in performance and are difficult to understand.

To solve these problems, this paper introduces a new model called the **Interpretable Spatial-Temporal Video Transformer (ISTVT)**. ISTVT uses a special decomposed self-attention mechanism that separately analyses spatial and temporal features. It also includes a "self-subtract" method to detect small changes between frames, helping to catch Deepfake signs more effectively.

In addition, the model offers clear visual explanations by highlighting the important regions in both space and time using a relevance propagation algorithm. Extensive testing on five popular Deepfake datasets: **FaceForensics++, FaceShifter, DeeperForensics, Celeb DF, and DFDC datasets** shows that ISTVT is both accurate and reliable. Its visual outputs also help users better understand how it makes decisions.

2. Introduction

2.1 Background on Deepfakes: Tools, Risks, and Real-World Impact

In the last few years, deepfake technology has grown very fast. Using tools like **Deepfakes**, **DeepFaceLab**, and **FaceSwap**, people can easily generate fake videos by swapping or editing faces. These videos are sometimes so realistic that it becomes hard to tell whether they’re real or fake. While this can be fun for entertainment, it can also be dangerous. Deepfakes can be misused to spread false information, attack

someone's reputation, or create fake evidence, which is a big threat to our privacy and security.
Because of this, researchers have been working on ways to detect these fake videos. There are already many deepfake detection systems, and they work well on big datasets like **FaceForensics++**, **Celeb-DF**, and **DFDC**. But as deepfake technology improves, these detectors are becoming less effective, especially when videos look very real or are in low quality.

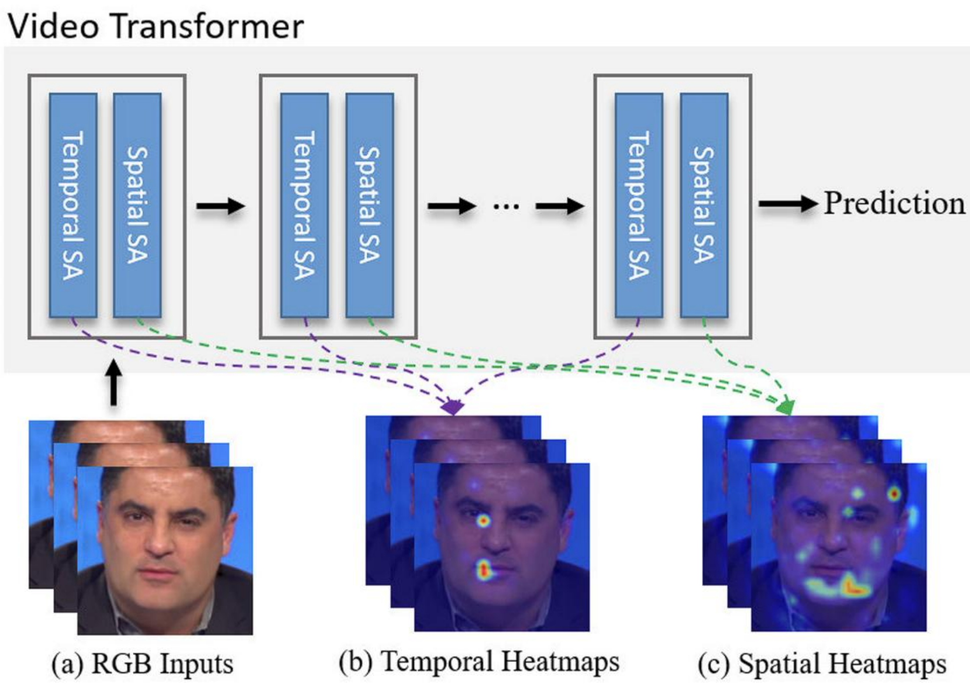
2.2 Need for Robust and Interpretable Detection

One big issue with many deepfake detectors is that they are not very robust, they can fail on new types of fakes or under different conditions. Also, they lack interpretability, meaning it's hard to understand why the model is saying a video is fake. As deepfakes become harder for even humans to catch, it becomes important for detection systems to not only be accurate but also explainable. We need models that can help us understand *how* they are making decisions, especially in critical areas like law, news, and online security.

2.3 Limitations of current detection methods

There are mainly two types of detection methods: **frame-based** and **video-based**.

- **Frame-based methods** look at single images (frames) from a video. They are good at finding small visual mistakes like blurred edges or strange textures. But they don't work well when:
    1. The deepfake is made using advanced techniques like NeuralTextures.
    2. The video is low quality or compressed.
    3. The method is trained on one type of deepfake and tested on another.
- To solve this, researchers have started using **video-based methods** that analyse a sequence of frames. These methods can spot **temporal inconsistencies**, weird movements or mismatches between frames. Since most fake videos are created frame-by-frame, they often mess up the consistency between frames. Video-based models aim to catch this.



Still, some of these models (like C3D and LSTM) have not given great results in practice. They also don't tell us what the model is focusing on when it decides a video is fake or real.

2.4 Rise of Transformer Models for Video and Their Challenges

Recently, **transformers** (like the ones used in language models) have been used for computer vision, including videos. Transformers can learn powerful patterns in data and have done well in tasks like **action recognition** and **video object detection**. Some researchers tried using transformers for deepfake detection, but most of them used the standard **vanilla vision transformer (ViT)**. These models are very **heavy** (need lots of computation) and still lack interpretability.

For example, in one such method (VTN), they applied normal vision transformers to video detection, but it ended up being slow and still not explainable.

2.5 Motivation for Building ISTVT

To solve the problems of poor robustness, lack of interpretability, and heavy computation, this paper introduces a new model called **ISTVT** (Interpretable Spatial-Temporal Video Transformer). It is specially designed for deepfake detection. ISTVT brings in two key ideas:

1. **Decomposed Spatial-Temporal Attention** – Instead of mixing both space and time, ISTVT separates them. This allows the model to learn both spatial features (like blurry eyes or fake skin) and temporal features (like unnatural movements) more effectively and clearly.
2. **Self-Subtract Mechanism** – This helps the model focus on what's changing between frames by subtracting features of adjacent frames. It removes unnecessary information and improves accuracy.

Additionally, ISTVT comes with a special **visualization method** that shows what the model is "looking at" in both space and time. This helps people understand how the model is making decisions.

3. Related Work / Literature Review

3.1 Video-Based Deepfake Detection

Early attempts to detect deepfakes were based on **hand-crafted features**, especially using **facial landmarks**. Researchers tried to find inconsistencies in landmark placements to detect if a face was fake or real. However, these methods were not very accurate because they depended a lot on how well the landmarks were detected, which is not always reliable.

Later, models that combined **CNN** (Convolutional Neural Networks) and **RNN** (Recurrent Neural Networks) were introduced. These aimed to capture both spatial and temporal features. But during the 2020 Deepfake Detection Challenge **(DFDC)**, this approach was found to be ineffective, and so it was not widely used afterward.

To improve results, some researchers used popular video models like **C3D**, **I3D**, and **LSTM**. These were general video analysis models, but they did not work very well for deepfake detection. In fact, their performance was often worse than simpler frame-based methods.

More recently, new models like **STIL** and **FTCN** were developed. STIL (Spatiotemporal Inconsistency Learning) aimed to detect both spatial and temporal artifacts in videos. FTCN (Fully Convolutional Temporal Network) used 3D CNNs and transformers to learn long-term temporal patterns. These approaches gave better results than older methods. But even with these improvements, they didn't focus on explaining what the model was learning separately in the temporal and spatial parts of the video.

One major issue with almost all previous methods is that they mix up spatial and temporal information in one step. This makes it hard to interpret what the model is actually focusing on is it looking at visual details in a single frame, or inconsistencies across frames?

3.2 Interpretability of Video Transformers

With the rise of transformers in computer vision, researchers began exploring their use for video deepfake detection too. Some basic interpretability methods like **raw attention scores**, **GradCAM**, and **Rollout** were used to visualize what these models were focusing on.

However, these methods are not ideal for transformers:

- Raw attention maps are often noisy and don't show clear explanations.
- GradCAM was designed for CNNs, so it doesn't work well with the token-based structure of transformers.
- Rollout shows average attention across layers but fails to separate spatial and temporal importance clearly.

One promising technique is Layer-wise Relevance Propagation **(LRP)**. LRP tries to explain how each part of the input contributes to the final decision. Some researchers used it for image transformers, and it showed much better visualization results.

But for video transformers, interpretability is still a challenge. Most methods don't separate the spatial attention (which parts of the image) from the temporal attention (which frame in the sequence). As a result, we don't get a clear picture of how the model is using the time-based information compared to image features.

This is where ISTVT stands out. It introduces a decomposed self-attention mechanism one part looks at spatial data and the other at temporal data. Because of this, ISTVT makes it possible to separately visualize and interpret what's happening over time and within each frame. Also, it builds upon the LRP method to provide accurate heatmaps for both dimensions.

This combination makes ISTVT not just effective in performance, but also a step forward in making deepfake detection more explainable and trustworthy.

4. Methodology

4.1 Overall Network Architecture

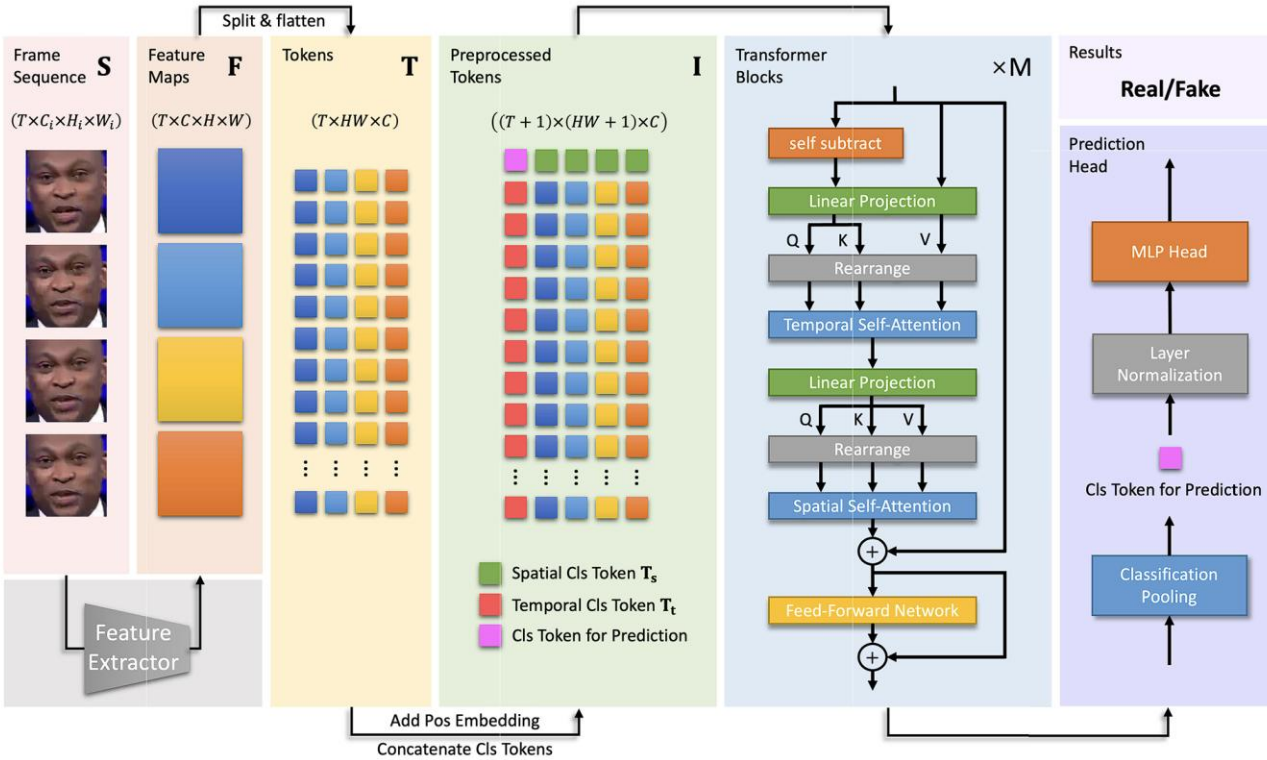


Fig: *The architecture of the proposed ISTVT*

The ISTVT methodology consists of four interconnected components that work together to achieve robust deepfake detection with interpretability. The architecture begins with feature extraction using an Xception backbone, followed by spatial-temporal transformer processing, and concludes with classification and interpretability analysis.

The network takes input video sequences of shape T × H × W × C, where T represents temporal length, H and W are spatial dimensions, and C denotes channels. The Xception feature extractor generates feature maps of shape T × H' × W' × C', which are then converted into patch tokens for transformer processing.

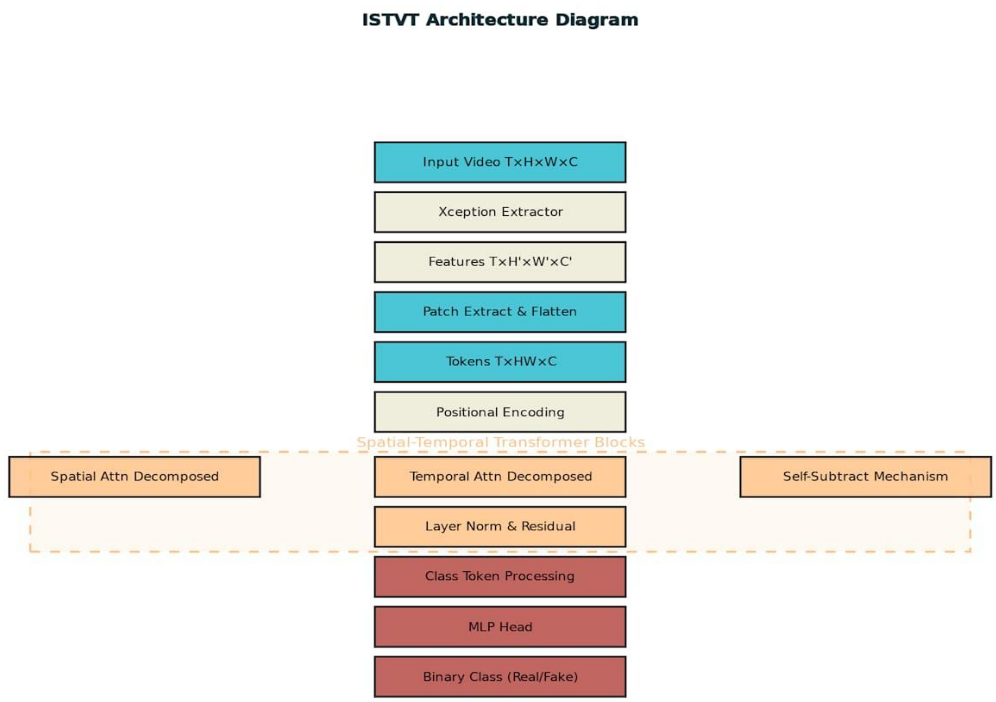


Fig: ISTVT Methodology Architecture - Detailed Block Diagram

The ISTVT model is made for video-based deepfake detection. It takes a sequence of video frames as input and processes them in several steps:

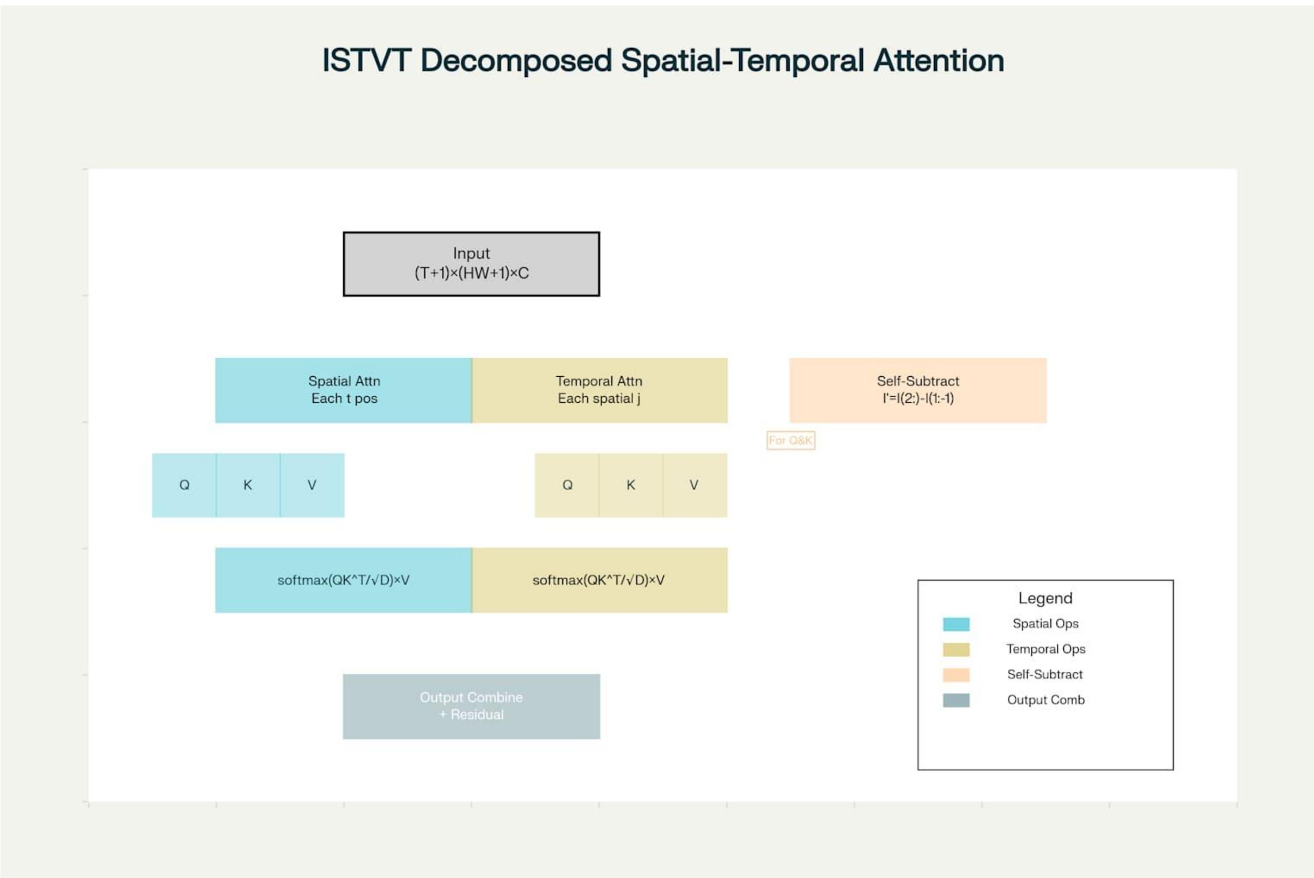
1. **Frame Sequence Input**:
   The input is a group of frames from a video (usually 6 frames), each with size 300×300. These are like small images showing different time points.
2. **Feature Extractor (Xception)**:
   Each frame is passed through a CNN model called Xception. This model is good at extracting low-level texture details, which is useful because deepfakes often leave behind texture flaws. The output is a feature map that highlights important visual patterns in each frame.
3. **Tokens**:
   The feature maps from all the frames are split into small patches (1×1) and flattened into tokens. Each token represents important information from a small area of the image.
4. **Transformer**:
   These tokens are fed into a Spatial-Temporal Transformer block that looks at

both the space (within a frame) and time (across frames). This helps the model find suspicious changes that occur over time.

5. **MLP Head**:
   The final output from the transformer is passed to a Multi-Layer Perceptron **(MLP)** classifier. This MLP predicts whether the input video clip is real or fake.

4.2 Spatial-Temporal Transformer Block

This is the main part of ISTVT. It uses decomposed self-attention, which means instead of looking at space and time together, it separates them:



1. **Spatial Self-Attention (within-frame)**:
   This checks for strange or fake-looking areas in one frame. For example, it may catch things like blurred skin, odd lighting, or weird facial boundaries.
2. **Temporal Self-Attention (across frames)**:
   This checks for changes between frames, like inconsistent movements of eyes, lips, or face structure. Deepfakes often fail to keep these things consistent.

By splitting the attention like this, the model can better understand and explain what it's focusing on. Also, this separation makes the computation faster and more memory-efficient compared to normal transformers that try to do both things at once.

4.3 Self-Subtract Mechanism

This is a unique idea in ISTVT. Before processing temporal information, the model subtracts one frame's features from the next. This does two things:

- It removes parts of the video that stay the same (like background, hair, etc.).
- It highlights the differences, which are often where the deepfake signs are (like glitches near the mouth or blinking).

In simple words, it's like asking: *"What has changed between these two frames?"* This helps the model focus on suspicious changes and ignore the parts that look normal, making the detection more accurate.

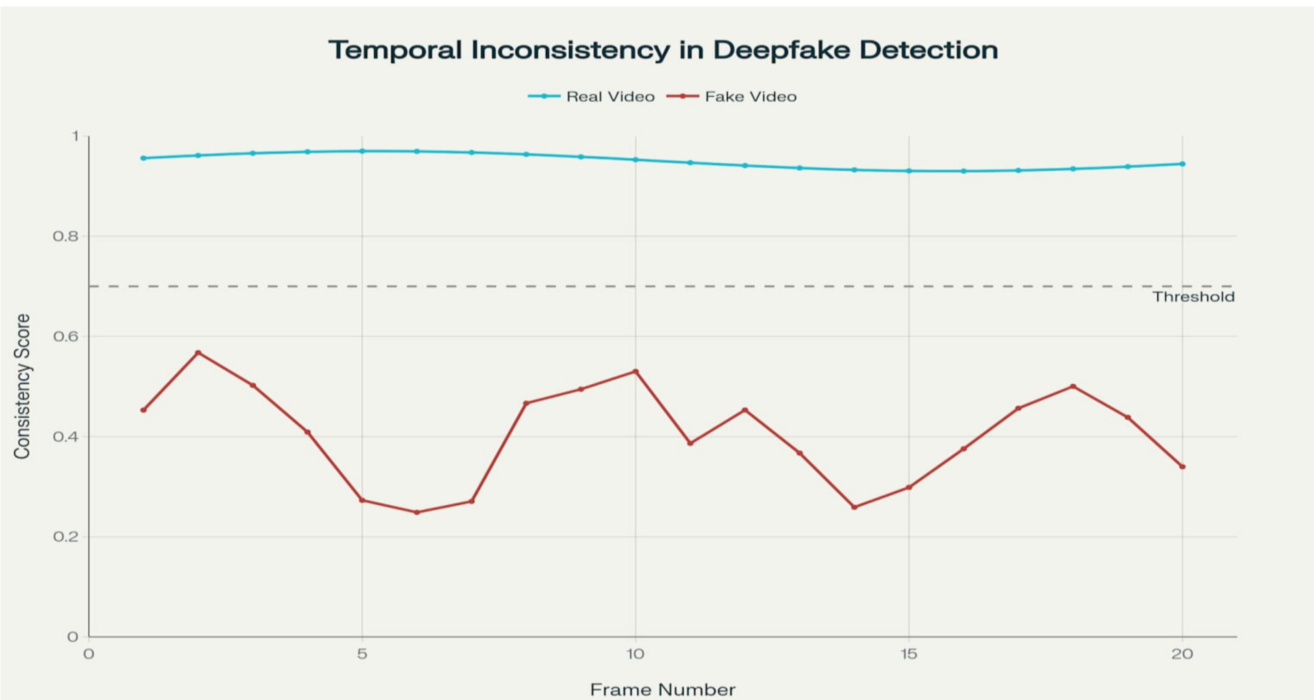


Fig : Temporal Inconsistency Analysis: Real vs Deepfake Videos

4.4 Model Interpretability

One of the coolest parts of ISTVT is that it is not a "black box" like many AI models. It can show what it's focusing on, using a method called Layer-wise Relevance Propagation **(LRP)**.

1. **LRP-based Relevance Heatmaps**:

   This generates two heatmaps:

- Spatial heatmap: Shows which part of the image the model thinks is fake.
    - **Temporal heatmap**: Shows which frames or parts of motion are suspicious.
2. **Why it's useful**:
   These heatmaps make the model explainable. For example, if the model says a video is fake, we can look at the heatmap and see it was because the lips were inconsistent across frames.
3. **Improves Trust**:
   When AI shows why it made a decision, people can trust it more. This is super important in areas like news, politics, or crime investigations.

5. Experiments

5.1 Datasets Used

To test how well the ISTVT model can detect deepfakes, the authors used five popular datasets. These datasets contain both real and fake videos, and each one has its own challenges. Using different datasets helps make sure the model works not just in one case, but in many different real-world situations.

1. FaceForensics++

FaceForensics++ (FF++) is one of the most commonly used datasets for deepfake detection. It includes:

- Real videos taken from YouTube.
- Fake videos generated using different face manipulation techniques like **Deepfakes**, **FaceSwap**, **Face2Face**, and **NeuralTextures**.

Each video comes in three levels of compression:

- **Raw (no compression)**,
- **HQ (high quality)**,
- **LQ (low quality)**.

This dataset helps test whether the model can still detect fakes even when the video is blurry or compressed. ISTVT uses all four manipulation types for training and testing.

2. FaceShifter

FaceShifter is a more recent and powerful face-swapping method. It creates very high-quality deepfakes that are hard to detect, even for humans.

- The dataset used here includes 500 fake videos created using FaceShifter and 500 matching real videos.
- All videos are compressed using H.264 codec, which is common on platforms like YouTube and Instagram.

Since these videos are high quality and realistic, they make the detection task more difficult. This helps to check if ISTVT can still catch very subtle and clean fakes.

3. DeeperForensics

DeeperForensics-1.0 is a large-scale dataset that is specially designed for real-world scenarios. It includes:

- 1000 real and 1000 fake videos.
- Videos are captured in varied lighting, background, and camera conditions.

This dataset is important because many models that work well in lab settings fail in real-world scenes. DeeperForensics helps test if ISTVT can handle real conditions like camera shake, lighting changes, and noise.

4. Celeb-DF

Celeb-DF (v2) contains videos of celebrities and is known for its very realistic fakes. It includes:

- 590 real videos and 5639 fake videos.
- The fake videos are generated using a refined deepfake synthesis method to avoid obvious flaws like flickering or bad mouth movements.

This dataset is very useful to test how the model performs when the deepfakes are extremely high quality and look almost real to the human eye.

5. DFDC (Deepfake Detection Challenge)

DFDC is one of the biggest and most challenging datasets. It was created by Facebook and includes:

- 19,154 videos from 4,113 real subjects.
- The dataset contains a wide variety of manipulations and face types.
- Many of the videos have low resolution, different angles, and realistic acting.

Because of its size and diversity, this dataset is used as a benchmark for many models. Testing on DFDC helps see whether ISTVT can generalize to new data and different fake creation styles.

5.2 Implementation Details

To train and test the ISTVT model properly, the researchers followed a detailed preprocessing and training setup. This helped ensure that the model was learning from clean, aligned face data and working efficiently with good performance.

Preprocessing Steps

Before giving the video frames to the model, the faces in each frame had to be detected and cleaned up. Here's how that was done:

1. **MTCNN (Multi-task Cascaded Convolutional Networks)**:
   This is a face detection tool. It was used to locate and extract faces from each video frame.
2. **Face Alignment**:
   After detecting the face, alignment was done to make sure all the faces are facing in the same direction. This helps the model compare facial features more accurately.
3. **Cropping**:
   Only the face region was cropped from the frame so that the background or unnecessary parts don't confuse the model.
4. **Resize**:
   The cropped faces were resized to 300 × 300 pixels, which is the fixed input size needed by the Xception feature extractor.

Training Setup

Once the frames were preprocessed, the model training began. The setup details are:

- **6-Frame Sequences**:
  Each video was broken down into chunks of 6 consecutive frames. These sequences act like short clips for the model to analyze.

- **Xception for Feature Extraction**:
  Each frame was passed through a pre-trained Xception network to extract useful features. These features were then turned into tokens for the transformer.
- **SGD Optimizer**:
  The model was trained using Stochastic Gradient Descent (SGD). This is a common method used to slowly adjust model weights and improve predictions.
- **Training Duration**:
  The model was trained for 100 epochs, which means the entire training data was shown to the model 100 times.

This setup allowed the ISTVT model to learn efficiently from both spatial and temporal patterns in video data, making it strong at detecting fake content.

5.3 Detection Performance

In this part of the experiments, the authors evaluate how well their ISTVT model can detect Deepfakes. They test it using two common methods in machine learning — intra-dataset and cross-dataset evaluations. These tests help us understand how accurate and general the model is in spotting fake videos.

1. Intra-Dataset Performance

In intra-dataset testing, the model is trained and tested on the same dataset. For example, the model is trained on FaceForensics++ (FF++) and also tested on FF++ itself. This setup checks if the model is learning Deepfake features effectively from the data.

- ISTVT achieved the highest accuracy on FF++, Celeb-DF, and DFDC datasets compared to other models.
- It was compared with models like Xception, ViViT, VidTr, and VTN. Most of them either use full-frame processing or general video transformers.
- Unlike these, ISTVT uses decomposed spatial-temporal self-attention, which lets it focus better on both spatial (within frame) and temporal (between frames) details.
- Even in difficult cases like low-quality videos or subtle manipulations, ISTVT performed better than traditional CNNs or existing video transformers.

This shows that ISTVT can detect Deepfakes more accurately when trained and tested on the same type of data.

2. Cross-Dataset Performance

Cross-dataset testing is more challenging. Here, the model is trained on one dataset (like FF++) but tested on different, unseen datasets like FaceShifter, DeeperForensics, Celeb-DF, and DFDC.

- This tests the model's generalization ability — how well it works on fake videos it has never seen before.
- ISTVT outperforms many strong models, including FTCN, LipForensics, and Face X-ray, on these unseen datasets.
- Especially on the DFDC dataset, which has more noise and diverse fake types, ISTVT showed better results than models like ViViT and VTN.
- One reason is that ISTVT focuses on short-term temporal inconsistency, which is more common in Deepfakes, rather than long-term patterns which can be misleading due to lighting or pose changes.

Conclusion of Performance Tests

From both the intra-dataset and cross-dataset experiments, we learn that:

- ISTVT is more accurate, more robust, and generalizes better than ISTVT is more accurate, more robust, and generalizes better than many other Deepfake detection models.
- Its unique design — especially decomposed attention and self-subtract — helps it detect both obvious and subtle Deepfakes in a smarter and more explainable way.

This strong performance proves ISTVT is a promising solution for real-world Deepfake detection problems.

5.4 Robustness Testing

In real-world scenarios, Deepfake detection models should not only be accurate but also robust. This means that even when videos are compressed, resized, or slightly distorted, the model should still be able to detect fakes correctly. In this section, the ISTVT model is tested against common video distortions to check its robustness.

The authors apply three types of perturbations to test the model:

1. JPEG Compression

JPEG compression is commonly used when videos or images are uploaded on social media or sent over the internet. But compression reduces the image quality by blurring details.

- In the experiment, the authors compressed test videos with different JPEG quality factors.
- ISTVT maintained higher accuracy than baseline models like Xception (a CNN) and VTN (a video transformer).
- This proves that ISTVT is more stable and less affected by blurry images, thanks to its decomposed spatial-temporal attention and self-subtract mechanism.

2. Downscaling

Downscaling means reducing the resolution of the video frames, which removes spatial details. But temporal movement between frames remains.

- As expected, frame-based models like Xception failed when the image size was too small.
- ISTVT performed much better because it focuses on temporal inconsistencies (movement differences) between frames, which stay even in low-res videos.
- This shows ISTVT is highly reliable even when image quality is poor.

3. Random Dropout

In this test, small square blocks (16×16) are randomly removed (dropped out) from the video frames to simulate noise or data loss.

- Frame-based models like Xception could not handle this well — they failed completely when 36 or more blocks were removed.
- ISTVT and VTN, being video-based models, handled it better.
- Especially ISTVT was able to still detect Deepfakes accurately because it relies on patterns across multiple frames, not just one.

Conclusion of Robustness Testing

Overall, these experiments prove that:

- ISTVT is significantly more robust than other models.

- It can detect Deepfakes correctly even when the input video is compressed, resized, or has missing data.
- Its special architecture, particularly the self-subtract mechanism, helps it focus on meaningful frame-to-frame differences, which stay even in noisy or low-quality videos.

This makes ISTVT highly practical and dependable for use in real-world Deepfake detection systems.

5.5 Ablation Study

The ablation study is a very important part of the experiments. It helps to understand which parts of the ISTVT model actually contribute the most to its high performance. In this study, the authors made changes to the model and checked how each change affected the results.

1. Comparing Different Attention Styles

The first test was about changing how the spatial-temporal self-attention is applied. Since ISTVT uses decomposed attention, the authors tried different orders:

- **Spatial-first**: Applies spatial attention before temporal.
- **Temporal-first**: Applies temporal attention before spatial.
- **Parallel**: Applies both at the same time using two separate branches.

The results showed that the temporal-first design gave the best performance. This means that first analysing how the face changes over time (between frames), and then looking at spatial features (within each frame), works best for Deepfake detection. This makes sense because many Deepfakes are made frame-by-frame, so spotting the inconsistencies over time first helps the model catch fakes more effectively.

2. Varying Model Depth (M) and Sequence Length (T)

Next, the authors tested different values for:

- M = number of transformer blocks (how deep the model is),
- T = number of video frames used in one input clip (sequence length).

They tried multiple values and found that:

- The best results came when T = 6 (6 frames per video clip).
- And M = 12 (12 transformer layers).

Using more frames (T > 6) didn't help much and even made training slower. Similarly, too many transformer layers (M > 12) increased the computation without improving accuracy.
So, T = 6 and M = 12 was the optimal choice, giving the best balance of accuracy and speed.

Conclusion of Ablation Study
The ablation study proves that:

- The temporal-first attention order works better than other versions.
- Choosing the right model depth and frame length is important to avoid unnecessary complexity.
- These design choices are not random — they are carefully selected based on real experiments.

This study helps to understand *why* ISTVT performs so well and proves that each part of the model design plays an important role in Deepfake detection.

6. Interpretability by Visualization

In Deepfake detection, it's not enough for a model to just give the right answer—we also want to understand why it made that decision. This is where interpretability becomes important. ISTVT is specially designed to not only detect Deepfakes but also to visually explain its predictions using attention heatmaps.
The authors use multiple visualizations to show how ISTVT makes decisions by separating spatial and temporal attention. These heatmaps allow us to see where (in space) and when (in time) the model is focusing.

6.1 Visual Results of Spatial vs Temporal Attention

Figures 9 to 12 in the paper show examples of these visual explanations. The results are very interesting:

- **Temporal Attention** focuses mostly on moving parts of the face, such as lips, eyes, and head motion. This is useful because Deepfakes often fail to maintain natural movements frame by frame.

- **Spatial Attention** highlights areas that have visual texture issues, such as blurry skin, artifacts, or inconsistent lighting. These are common problems in generated images.

By separating these two attentions, ISTVT allows us to see exactly what the model is looking at, and whether it's detecting motion inconsistencies or texture problems.

6.2 Challenging Examples and the Role of Self-Subtract

One of the most impressive parts of the visualization is how the self-subtract mechanism helps correct wrong predictions.

For example, in one difficult case shown in the paper, the model without self-subtract wrongly predicted a real video as fake. But with the self-subtract module, the model ignored the static background and focused only on the face motion, leading to a correct prediction.

This proves that the self-subtract layer helps the model remove unnecessary information and focus on the most relevant parts of the video—especially changes across frames, which are key to catching Deepfakes.

6.3 Comparison with Other Visualization Methods

The paper also compares ISTVT's attention heatmaps with those generated by GradCAM, Rollout, and Raw Attention Scores.

- **GradCAM** was originally made for CNNs, so it doesn't work well with transformers.
- **Rollout** shows average attention, but it mixes up spatial and temporal signals, making it hard to interpret.
- **Raw Attention Scores** are often noisy and not very informative.

In contrast, ISTVT uses Layer-wise Relevance Propagation (LRP) with decomposed attention. This provides cleaner and more meaningful visualizations, clearly showing both spatial and temporal focus areas.

6.4 Perturbation-Based Accuracy Test

To test whether these attention maps are not just pretty pictures but actually meaningful, the authors performed a perturbation test.

In this test:

- They removed the regions with the highest attention (as shown by the heatmaps).
- If the model's prediction changes a lot, it means the heatmap pointed to something important.

Results showed that ISTVT's attention maps caused the most change in predictions when removed, proving that:

- The model really relies on these regions for making decisions.
- ISTVT's interpretability is more accurate and reliable than other methods.

7. Conclusion and Future Work

In this report, we studied the ISTVT model – Interpretable Spatial-Temporal Video Transformer – which is designed to detect Deepfake videos more accurately and explainably. Through all the experiments and tests, we found that ISTVT performs very well in terms of accuracy, robustness, and interpretability.

One of the main strengths of ISTVT is that it separately models spatial and temporal inconsistencies. This means it looks at both:

- **Spatial artifacts** like blurred textures and unnatural face edges within a single frame.
- **Temporal inconsistencies** like strange lip or eye movements across multiple frames.

This decomposed self-attention mechanism helps the model understand fake videos better and also reduces computation cost. Also, the self-subtract mechanism allows the model to focus on only the changing parts of the video, which usually contain Deepfake clues.

Another important contribution of ISTVT is its visualization ability. By using Layer-wise Relevance Propagation (LRP), the model generates clear attention maps that show where and when it focuses in a video. This helps users and researchers understand the model's decisions and builds trust in its predictions.

Future Work

While ISTVT already shows strong results, the authors suggest a few directions for future improvement:

- They plan to work on better visualization techniques to make model interpretation even more accurate and user-friendly.
- ISTVT could also be adapted for other video analysis tasks, such as action recognition or emotion detection, where spatial-temporal understanding is important.

Overall, ISTVT is a big step forward in making Deepfake detection not only powerful but also interpretable, explainable, and reliable for real-world use.

8. References

This report is primarily based on the research article:

Zhao, C., Wang, C., Hu, G., Chen, H., Liu, C., & Tang, J. (2023). **ISTVT: Interpretable Spatial-Temporal Video Transformer for Deepfake Detection.** *IEEE Transactions on Information Forensics and Security, Vol. 18*, pp. 1335–1348.
https://doi.org/10.1109/TIFS.2023.3239223

All the technical explanations, figures, methodology, experiments, and analysis presented in this report have been thoroughly summarized and interpreted from the above paper. Additional references cited within that research work (total 59) cover datasets, model architectures, deepfake tools, and related prior methods. For the complete list, kindly refer to the Reference section of the original IEEE paper.
